# A chiral coset geometry with disconnected residues

Human Compression project

**Abstract**

We give a rank-four counterexample to the stated residual-connectedness criterion for chiral geometries constructed from $C^+$-groups. The specified group is $\mathrm{PSL}_2(13)$, whereas the full type-preserving automorphism group is $\mathrm{PSL}_2(13) \times C_2$. The geometry has 112 objects and 4368 chambers. Its full group has two chamber orbits, with adjacent chambers in different orbits, but a rank-two residue is the disjoint union of two six-cycles. We prove the full automorphism assertion by an elementary calculation on thirteen points; no automorphism-group software or classification is required.

## 1 The decorated point graph

All entries in this note are in $\mathbb{F} = \mathbb{F}_{13}$. Let

$$K = \{1, 3, 4, 9, 10, 12\} = (\mathbb{F}^\times)^2, \qquad V = (\mathbb{F}^2 \setminus \{0\})/K.$$

Write $[v]$ for the class $Kv$. There are 28 classes, two on each projective line. Define a simple undirected graph $X$ on $V$ by

$$[v] \sim [w] \quad \Longleftrightarrow \quad \det(v, w) \in K. \tag{1}$$

The condition is independent of the representatives and symmetric because $-1 \in K$. Points on the same projective line are not adjacent. Put $x = [(1, 0)^T]$, $y = [(0, 1)^T]$, and $z = [(1, 1)^T]$.

The natural $\mathrm{SL}_2(13)$ action preserves (1). Its kernel is $\{\pm I\}$: a matrix fixing $x, y, z$ must be scalar, and a determinant-one scalar has square one. Hence its image $G = \mathrm{PSL}_2(13)$ has order $13(13^2-1)/2 = 1092$. It is transitive on vertices and on ordered edges. Indeed a nonzero vector can be completed to a determinant-one basis; for an adjacent ordered pair one can first rescale one vector by a member of $K$ to obtain the required determinant. Every vertex has thirteen neighbours: those of $x$ have unique representatives $(a, 1)^T$, $a \in \mathbb{F}$. The graph induced on this neighbourhood is

$$a \sim b \quad \Longleftrightarrow \quad a - b \in K, \tag{2}$$

the Paley graph on thirteen points. In particular every edge of $X$ has six common neighbours.

Multiplication of every vector by 2 induces an automorphism $s$ of $X$, since $4 \in K$. It is an involution and commutes with $G$. It does not belong to $G$. Otherwise a determinant-one matrix inducing this action would be $2kI$ for some $k \in K$, forcing $2k = \pm 1$, whereas $\pm 7 \notin K$. Thus

$$A = \langle G, s \rangle \cong G \times C_2, \qquad |A| = 2184. \tag{3}$$

Every element of $A$ is induced by an invertible linear matrix, taken modulo the scalar subgroup $K$.

**Lemma 1.** *The full automorphism group of $X$ is $A$.*

*Proof.* The other decorated point on the line of $[v]$ is $[2v]$. For each different line, exactly one of its two decorations is adjacent to $[v]$, and the other is adjacent to $[2v]$. These two vertices therefore have no common neighbour. Two vertices on distinct lines do have a common neighbour: solve the independent linear equations $\det(v, w) = \det(v', w) = 1$. Thus $[2v]$ is the unique other vertex with no common neighbour with $[v]$, and every automorphism preserves this antipodal pairing.

An automorphism fixing $x$ and its thirteen neighbours fixes their antipodes and the antipode of $x$, hence every vertex. Consequently the stabilizer of $x$ embeds in the automorphism group of (2). We bound the latter group directly. The neighbours of 0 in (2) induce the six-cycle, in the following cyclic order:

$$(1, 4, 3, 12, 9, 10). \tag{4}$$

Number its positions $0, \ldots, 5$. The six remaining nonzero vertices have the following neighbour sets on this cycle:

| vertex | 2 | 5 | 6 | 7 | 8 | 11 |
|---|---|---|---|---|---|---|
| positions | 023 | 014 | 245 | 125 | 134 | 035. |

(5)

The six triples in (5) are the translates of $\{0, 2, 3\}$ in $\mathbb{Z}/6\mathbb{Z}$. They are distinct, so an automorphism fixing 0 is determined by its action on (4). This action belongs to the dihedral group of the cycle. Every rotation preserves the collection (5), while the reflection $i \mapsto -i$ sends $\{0, 2, 3\}$ to $\{0, 3, 4\}$, which is absent. No reflection preserves the collection. The stabilizer of 0 in the automorphism group of (2) therefore has order at most six, and that group has order at most $13 \cdot 6 = 78$.

It follows that $|\operatorname{Aut}(X)| \leq 28 \cdot 78 = 2184$. The subgroup (3) attains the bound, proving equality. □

There are $28 \cdot 13 \cdot 6 = 2184$ ordered triangles in $X$. An element of $A$ fixing one of them fixes three different projective lines and is scalar; fixing the decorated points then makes it the identity modulo $K$. Thus $A$ acts freely and transitively on the ordered triangles. In particular each triangle has the same number of common neighbours as $x, y, z$. A common neighbour of $x, y$ is represented by $(a, 1)^T$ with $a \in K$. It is adjacent to $z$ precisely when $1 - a \in K$. Directly from the six displayed elements of $K$, the solutions are $a = 4, 10$. Every triangle has exactly two common neighbours.

## 2 The geometry and its full chirality

Let $I = \{0, 1, 2, 3\}$. Take four tagged copies of $V$, and declare objects of different types incident precisely when their underlying vertices are adjacent in $X$. Equality is the only incidence within one type. Denote this incidence system by $\Gamma$. Every incident pair extends to a triangle; every triangle has exactly two common neighbours. Hence every flag extends to a chamber and every panel has exactly two completions. Thus $\Gamma$ is a thin geometry. Its objects number $4 \cdot 28 = 112$, and its chambers number $2184 \cdot 2 = 4368$.

**Lemma 2.** *The full type-preserving automorphism group of* $\Gamma$ *is the diagonal action of* $A$*. It has exactly two chamber orbits, and adjacent chambers belong to different orbits.*

*Proof.* First, distinct vertices of $X$ have distinct neighbourhoods. Antipodes have disjoint neighbourhoods. If $u, v$ are on different lines, one of the two decorations on the line of $u$ is adjacent to $v$, while neither is adjacent to $u$; their neighbourhoods therefore differ.

Write a type-preserving automorphism as permutations $f_i$ on the four copies of $V$. For distinct $i, j$, incidence gives $N(f_i(v)) = f_j(N(v))$. Given $i, k$, choose $j$ distinct from both. The preceding equality gives $N(f_i(v)) = N(f_k(v))$, so the absence of equal neighbourhoods

implies $f_i = f_k$. All four permutations are the same automorphism of $X$. Lemma 1 proves the first assertion.

This group is free on chambers, since it is free on ordered triangles. Its order and the chamber count give exactly two orbits. If an automorphism took one chamber to an adjacent chamber, it would fix the three common objects and hence be the identity. Adjacent chambers are distinct, so this cannot happen. This is chirality under the full type-preserving group, not merely under a proposed subgroup. □

The geometry is not residually connected. Fix the type-1 object $x$ and the type-2 object $y$. The remaining objects in their residue are two copies, of types 0 and 3, of

$$\{[(a,1)^T] : a \in K\}.$$

Their incidence is $a - b \in K$, so this residue is the bipartite double of the cycle (4). Since that cycle has even length, its bipartite double is two disjoint six-cycles: one component uses the even cycle positions in type 0 and the odd positions in type 3, and the other uses the opposite positions.

## 3 The exact $C^+$ coset construction

It remains to verify the algebraic hypothesis in the source theorem. In $G = \mathrm{PSL}_2(13)$ use the determinant-one matrices

$$\alpha_0 = I, \qquad \alpha_1 = \begin{pmatrix} 4 & 0 \\ 7 & 10 \end{pmatrix}, \quad \alpha_2 = \begin{pmatrix} 4 & 6 \\ 0 & 10 \end{pmatrix}, \quad \alpha_3 = \begin{pmatrix} 3 & 0 \\ 0 & 9 \end{pmatrix}. \tag{6}$$

Choose the following base objects, in type order:

$$v_0 = [(4,1)^T], \qquad v_1 = x, \qquad v_2 = y, \qquad v_3 = z. \tag{7}$$

They form a chamber, as their six determinants belong to $K$. For $S \subseteq I$ put $P_S = \langle \alpha_i^{-1}\alpha_j : i,j \in S \rangle$. The six differences for $i < j$ are as follows; the last column gives their two fixed base objects.

| $ij$ | $\alpha_i^{-1}\alpha_j$ | fixed labels |
|---|---|---|
| 01 | $\left(\begin{smallmatrix} 4 & 0 \\ 7 & 10 \end{smallmatrix}\right)$ | $2,3$ |
| 02 | $\left(\begin{smallmatrix} 4 & 6 \\ 0 & 10 \end{smallmatrix}\right)$ | $1,3$ |
| 03 | $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 9 \end{smallmatrix}\right)$ | $1,2$ |
| 12 | $\left(\begin{smallmatrix} 1 & 8 \\ 11 & 11 \end{smallmatrix}\right)$ | $0,3$ |
| 13 | $\left(\begin{smallmatrix} 4 & 0 \\ 5 & 10 \end{smallmatrix}\right)$ | $0,2$ |
| 23 | $\left(\begin{smallmatrix} 4 & 11 \\ 0 & 10 \end{smallmatrix}\right)$ | $0,1$ |

(8)

Each is non-scalar and its cube is $I$ or $-I$, so it has order three in $G$. Each fixes the two displayed decorated points, not just their projective lines. No nonidentity projective transformation fixes three distinct lines.

The stabilizer in $G$ of any decorated point is $C_{13} : C_3$, of order 39. For $x$ this follows directly from its upper triangular lifts, whose first diagonal entry belongs to $K$; the others follow by transitivity. For each $i$, (8) puts $P_{I\setminus\{i\}}$ in this stabilizer of $v_i$ and supplies two distinct subgroups of order three, distinguished by their second fixed lines. A subgroup of $C_{13} : C_3$ containing two such subgroups must be the whole group: if it met the normal $C_{13}$ trivially it would embed in $C_3$, and a nontrivial intersection is the whole $C_{13}$. Thus

$$H_i := P_{I\setminus\{i\}} = G_{v_i}, \qquad |H_i| = 39. \tag{9}$$

The three matrices generate $G$. Indeed $H_1$ contains every upper unipotent matrix and $H_2$ every lower unipotent matrix. These generate $\mathrm{SL}_2(13)$ before passing to its projective quotient. For instance the explicit products

$$\alpha_2\alpha_3 = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}, \qquad \alpha_1\alpha_3 = \begin{pmatrix} -1 & 0 \\ 8 & -1 \end{pmatrix}$$

already give nonzero upper and lower translations, whose powers give all such translations. Row elimination then gives every determinant-one matrix.

The intersection property is now explicit. Two distinct groups in (9) intersect in a group of order three, since $G$ is transitive on the $28 \cdot 13$ ordered edges. Their intersection is the group $P_S$ from (8) on their two complementary labels. Three distinct groups in (9) intersect trivially. A cyclic $P_S$ with $|S| = 2$ is contained in $H_i$ exactly when $i \notin S$; otherwise an element of the intersection would fix three base lines. Two different such cyclic groups likewise intersect trivially. Together with $P_I = G$ and $P_S = 1$ for $|S| \leq 1$, these observations prove

$$P_S \cap P_T = P_{S \cap T} \qquad (S, T \subseteq I). \tag{10}$$

The generators are independent: omitting $\alpha_i$, $i > 0$, leaves $H_i$ of order 39. Hence (6) is a $C^+$-marking in the source's exact convention.

Finally the source's right coset $H_i g$ corresponds to $g^{-1}v_i$. Equality of fibres follows from (9). Coset incidence is precisely the $G$-orbit of the corresponding base pair; every base pair in (7) is an edge, and $G$ is transitive on all ordered edges. Therefore all six coset incidence relations agree with (1). This proves that $\Gamma$ is exactly Construction 3.2 for (6), rather than a geometry which merely contains a suitable group action.

**Theorem 3.** *The $C^+$-group (6) has a chiral coset geometry that is not residually connected. Its full type-preserving group is $\mathrm{PSL}_2(13) \times C_2$, its specified group is $\mathrm{PSL}_2(13)$, and its rank-two residue at the type-1 and type-2 objects of (7) is two disjoint six-cycles.*

## 4 Source scope and verification

The theorem contradicts Theorem 1.1 in the fixed published edition of Leemans–Tranchida [1]. Their definition on printed page 493 uses the full $\mathrm{Aut}_I(\Gamma)$ to define chirality; Lemma 2 verifies exactly that definition. Construction 3.2 on page 494 is paid by (6)–(10) and the coset identification. The geometry is thin and every flag extends, so neither of these hypotheses is missing.

The issue is the passage on printed page 496 from proper-flag transitivity under the full group to transitivity under the specified $G$. Here the full group acts freely and transitively on flags of any three fixed types, while $G$ has two orbits on those flags. The internal rank-two coset model thus covers only one of the two actual residue cycles. This counterexample does not refute the separately valid rank-three conclusion in Proposition 4.11, nor a criterion which explicitly requires the specified group to have the two alternating chamber orbits. It is a correction of a published theorem, not a claim to have solved a named open conjecture.

The decorated-vector vocabulary comes from the accompanying recovery of the tetrahedral construction of Catalano, Fernandes, Hubard and Leemans [2]. That source uses point and tetrahedron types. The present construction instead takes four point types. All group and incidence arguments needed for this example are given here. We make no claim of priority for the graph (1), the Paley graph, or their standard matrix actions.

The accompanying short program independently enumerates the 28 vertices, all triangles and four-cliques, the generated 1092-element matrix group and all 256 intersections in (10). It checks the full ordered-triangle action, both chamber orbits, the six local attachment triples and the disconnected actual residue. These are finite checks of the displayed example; the full automorphism upper bound is the written argument of Lemma 1.

## References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics 4 (2021), 491–499, doi:10.5802/alco.162.

[2] D. Catalano, M. E. Fernandes, I. Hubard and D. Leemans, *Hypertopes with tetrahedral diagram*, Electronic Journal of Combinatorics 25(3) (2018), P3.22.